WEB TRAFFIC ANALYSIS USING CLOUDFLARE API

Presented By

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Cloudflare

Web security & performance platform

Filters malicious traffic

Monitors human and bot interactions

It improves website reliability and speed

API

Secure communication bridge

Allows data retrieval

Uses authentication token

API token ensures authenticated access

Data Collection

Generated API token from Cloudflare

Extracted traffic data

API with Python in Google Colab

Retrieved the response into DataFrame

Feature

Original

- Features
- Timestamp
- Human Traffic
- Bot Traffic

Derived

- Traffic Difference
- Activity Classification of bot
- Traffic Status
- Trend & Spike Indicators

OBJECTIVE

To differentiate human and bot traffic patterns.

To analyze traffic variation across time and activity levels.

To evaluate traffic health status. To detect abnormal spikes and trend shifts.

To derive actionable insights using data analysis techniques.

Exploratory Data Analysis

Verified dataset structure using df.info()

Examined dataset dimensions using df.shape()

Generated statistical summary using df.describe()

Checked for null values and duplicate records

Confirmed data consistency before analysis

Statistical Summary

	Human	Bot	HumanTraffic Ratio	BotTraffic Ratio	Traffic Difference	Bot Spike	Bot Change
mean	70.326284	29.673716	0.703263	0.296737	40.652567	0.447989	0.001528
min	64.522657	25.786139	0.645227	0.257861	29.045314	0.000000	-4.197480
max	74.213861	35.477343	0.742139	0.354773	48.427722	1.000000	3.050463
std	1.897955	1.897955	0.018980	0.018980	3.795911	0.497633	0.629766

Central Tendency

Human traffic average (~70%) is consistently higher than bot traffic (~30%), showing dominant genuine user activity.

Stability & Spread

Low standard deviation values indicate minimal fluctuation and stable traffic behavior.

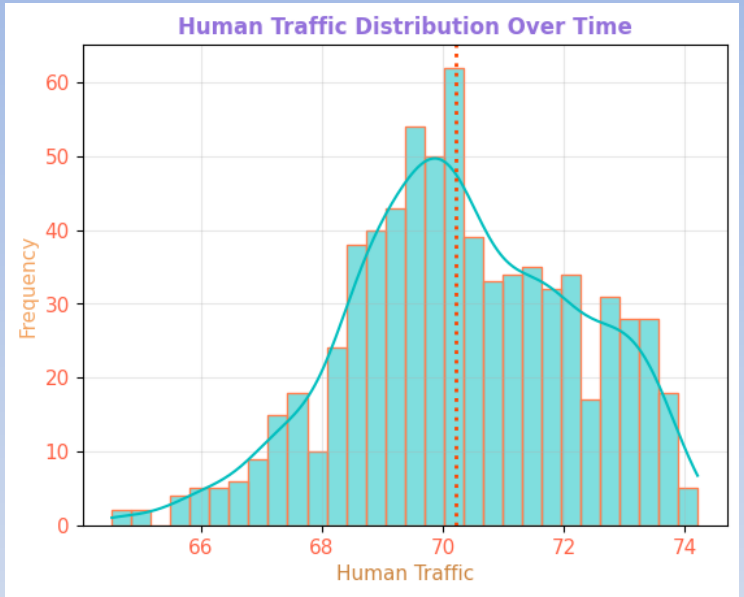
Traffic Balance

Human and bot traffic ratios reflect a balanced distribution without extreme dominance.

Controlled Variation Traffic difference and bot spike metrics show manageable variation, indicating no severe anomalies.

Analytical Framework	Univariate	Bivariate	Multivariate
Multi-level traffic behavior analysis	Analysis of in dividual traffic variables	Examines relationship between two variables	Studies interaction of multiple variables
	Examines distribution and variation	Identifies correlation patterns	Analyses traffic behaviour across time & activity levels
	Identifies dominant traffic patterns	Compares human vs bot interactions	Provides deeper behavioural insights

Univariate Analysis



Distribution Over Time

Human traffic follows a near-normal distribution, with most values concentrated around the 70% range.

The distribution shows moderate variation, indicating stable and consistent human engagement over time.

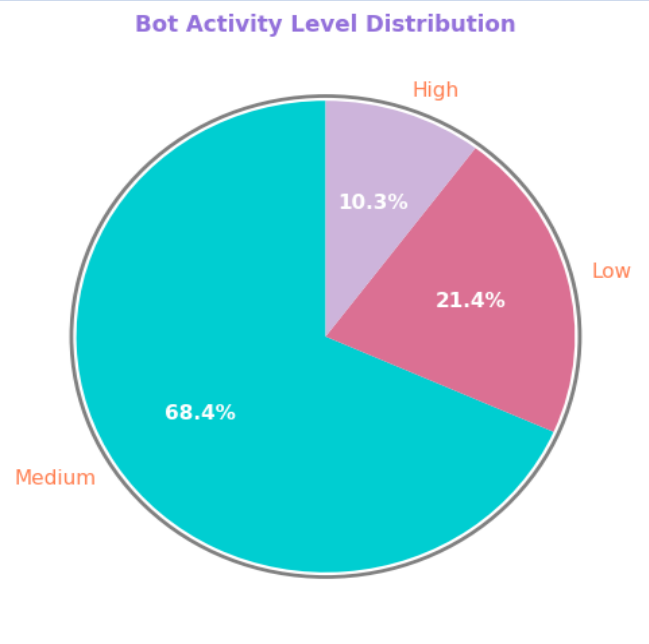
There are no extreme outliers, which suggests predictable human traffic behavior.

Activity Level Distribution

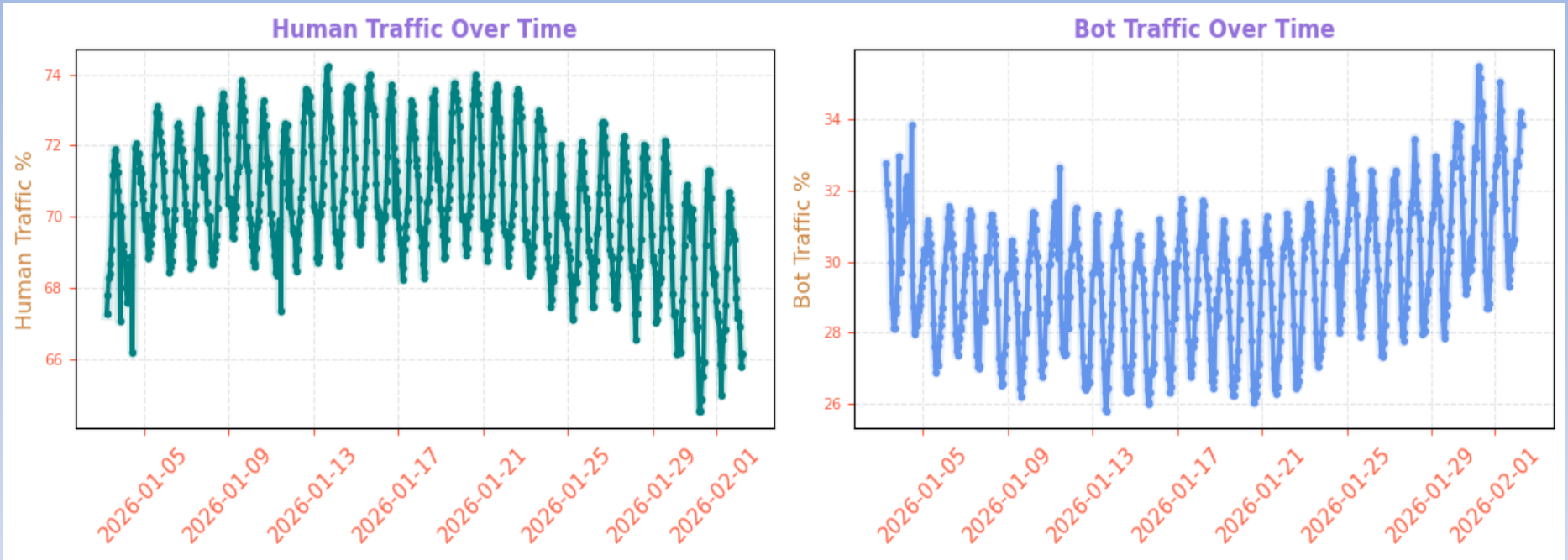
The majority of bot activity falls under the “Medium” category, indicating controlled automated traffic.

High bot activity represents a smaller portion, but it is critical for anomaly detection and monitoring.

Low bot activity indicates normal background automation without significant threat levels.



Bivariate Analysis



Human traffic shows a consistent cyclical pattern, indicating predictable user behavior over time.

The overall trend remains relatively stable with moderate fluctuations, suggesting steady engagement.

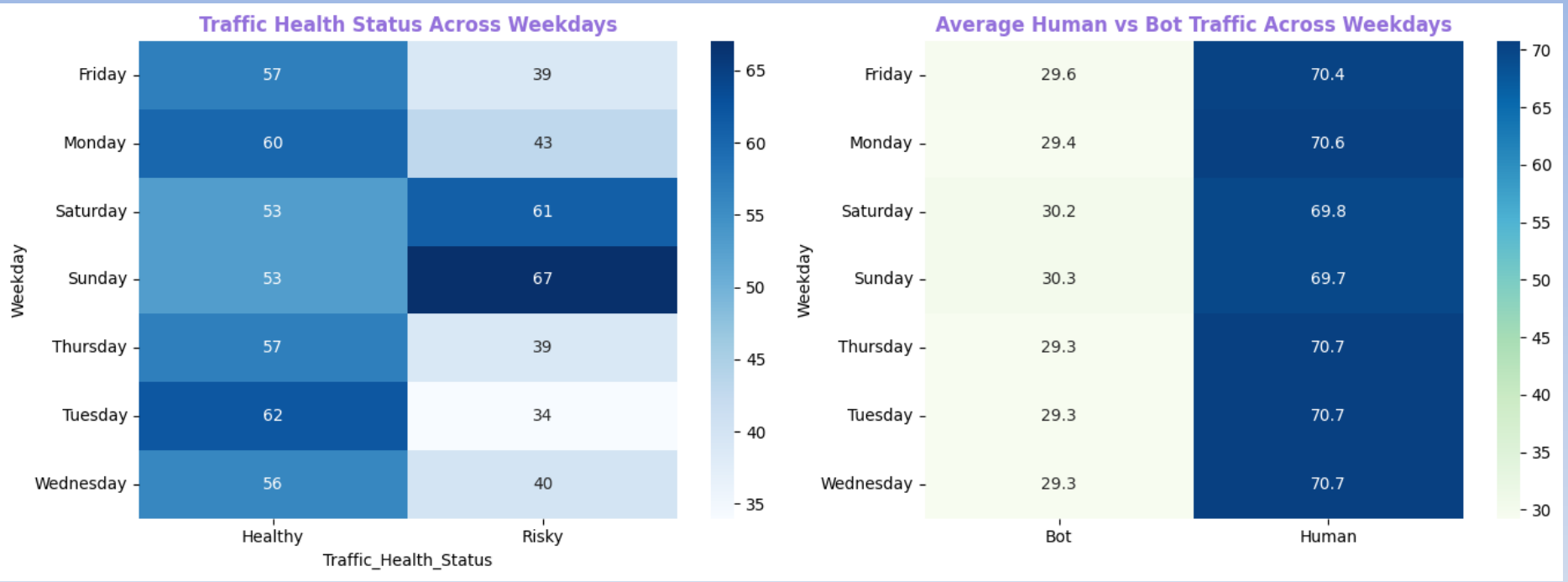
There are no extreme sudden drops, which indicates healthy and reliable human interaction.

Bot traffic shows more variability compared to human traffic, indicating automated activity fluctuations.

There is a noticeable upward trend toward the later period, which may indicate increasing bot intensity.

Certain peaks suggest abnormal bot spikes that require monitoring for potential security risks.

Multivariate Analysis



Traffic health varies across weekdays, indicating that user engagement and bot influence fluctuate depending on the day.

Certain days show higher “Healthy” status, suggesting stronger human dominance and stable website performance.

Days with relatively lower health scores may indicate increased bot activity or abnormal traffic behavior.

Human traffic consistently remains higher than bot traffic across all weekdays.

There is slight variation in averages, showing weekday-based traffic behavior differences.

The stable gap between human and bot traffic indicates controlled automated activity

Conclusion:

Human traffic consistently dominates bot traffic (~70% vs ~30%), indicating healthy user engagement.

Traffic distribution remains stable with low variability, confirming controlled website activity.

Bot activity shows moderate fluctuations but no extreme anomalies or sustained abnormal spikes.

Correlation and multivariate analysis indicate predictable traffic behavior across time and activity levels.

Overall, the website traffic ecosystem appears stable, balanced, and operationally healthy.

Future Enhancement:

Implement real-time monitoring dashboard for automated bot spike detection.

Integrate predictive modeling to forecast high bot activity periods.

Apply anomaly detection algorithms for enhanced security insights.

Extend analysis to longer time ranges for deeper behavioral trends.

Combine traffic analysis with geographic or device-level data for richer insights.